Millimeter-Wave Substrate Integrated Waveguides and Filters in Photoimageable Thick-Film Technology

Daniel Stephens, Paul R. Young, *Senior Member, IEEE*, and Ian D. Robertson, *Senior Member, IEEE*

***Abstract*—This paper presents the design and fabrication of substrate-integrated waveguides and filters for use at millimeter-wave frequencies. The components described are fabricated using photoimageable thick-film materials. Measurements on V- and W-band waveguide-to-microstrip transitions are presented. Losses resulting from the reduced-height nature of the waveguides are extracted from thru-relect-line calibration standards illustrating the effect of current losses in the broadwalls of the waveguides. The design of fourth-order 0.01-dB ripple Chebyshev resonant cavity filters operating at V-, W-, D-, and G-band is presented. The measured results of all components are in excellent agreement with simulated predictions.**

***Index Terms*—Ceramics, filters, millimeter waves, substrate integrated waveguides (SIWs), thick-film technology.**

I. INTRODUCTION

IN RECENT years, much interest has developed in systems operating in the millimeter-wave regime. This interest is largely based on the unique spectral characteristics of the atmosphere, affecting the propagation of electromagnetic waves above 10 GHz [1]. The presence of absorption lines and transmission windows within the atmosphere has given way to the development of systems used in a wide variety of applications. These include remote sensing of the H_20 absorption lines (occurring at 22 and 183 GHz) for the monitoring of climate change, passive millimeter-wave imaging (at 94 GHz), capable of all-weather operation, and also the development of wireless communication networks (at 60 GHz), which take advantage of the increased atmospheric attenuation (10–15 dB km^{-1}) of the O_2 line to provide for frequency reuse.

A practical solution for millimeter-wave transceiver modules required for these applications consists of a system-in-package (SiP) approach with Si, GaAs, and InP devices integrated on a single substrate with embedded passive components. However, the use of conventional planar transmission lines within a SiP results in significant attenuation when operating above 100 GHz. Considerable research has thus been reported on methods of fabricating rectangular waveguides (RWGs) using photolithographical methods. RWGs have excellent power-handling capabilities, are well understood, and are immune from radiation loss and crosstalk. In addition, the attenuation and dispersion characteristics of RWGs can be accurately modeled mathematically or with the aid of three-dimensional electromagnetic simulators. Moreover, a vast array of waveguide components and designs are available to the microwave engineer, including high-Q filters, compact antennas, power combiners, and broad-band couplers.

However, before the advantages of this medium can be drawn upon, the manufacture of these waveguides needs to be accomplished with sufficient accuracy so as to allow for operation at millimeter-wave frequencies. At very high frequencies, micromachining has been successfully used to fabricate hollow waveguides [2], [3]. For ease of integration with other components, the substrate integrated waveguide (SIW) has attracted much interest. Here, the waveguide is dielectric filled and embedded into a substrate, which can also be used for microstrip and CPW lines, lumped elements, and as a chip carrier for integrated circuits. Hence, the SIW is a very promising approach for realizing a millimeter-wave multichip module [4]. The methods that have been used to realize SIWs include multilayer GaAs monolithic microwave integrated circuit (MMIC) [5], low-temperature co-fired ceramic (LTCC) [6], standard microwave integrated circuit (MIC) [7], and thick-film [8] technology. It has been shown that a wide range of components can be integrated including filters [9]–[13], antennas [14], [15], phase shifters [16], and circulators [17].

In LTCC technology, the continuous sidewalls are usually replaced with a periodic row of metallized via-holes, which is more compatible with the processing of the tapes. The propagation and leakage characteristics of a guide with this structure have been studied [18]. The size of the via-holes is quite large in LTCC and this limits the technique currently to the lower end of the millimeter-wave spectrum. The minimum via-to-via spacing requirement results in increased radiation losses at higher millimeter-wave frequencies. Multilayer GaAs MMIC fabrication allows solid sidewalls to be formed. However, the waveguides in [5] suffered from very high losses due to fabrication limitations on the waveguide height.

The multilayer thick-film approach to fabrication of SIWs has the advantage that continuous metal sidewalls can be fabricated and low-loss dielectric pastes are available, including materials from Heraeus [19], Dupont [20], and Hibridas [21]. The Hibridas photoimageable technology has previously been shown to be suitable for realizing SIWs up to at least 100 GHz with excellent dimensional tolerances and low dielectric loss. Transitions, filters, and antennas have previously been reported using this technology [8], [12], [15]. With thick-film fabrication, the challenge is that highly three-dimensional structures are difficult to print with high resolution and, thus, waveguide height is limited. This paper, therefore, describes the fabrication process and its limitations, the propagation characteristics

Manuscript received May 3, 2005.

D. Stephens is with the Microwave Systems Research Group, University of Surrey, Guildford, Surrey GU2 7XH, U.K.

P. R. Young is with the Department of Electronics, University of Kent, Canterbury, Kent CT2 7NT, U.K. (e-mail: p.r.young@kent.ac.uk).

I. D Robertson is with the Institute of Microwaves and Photonics, The University of Leeds, Leeds LS2 9JT, U.K. (e-mail: i.d.robertson@leeds.ac.uk).

Digital Object Identifier 10.1109/TMTT.2005.859862

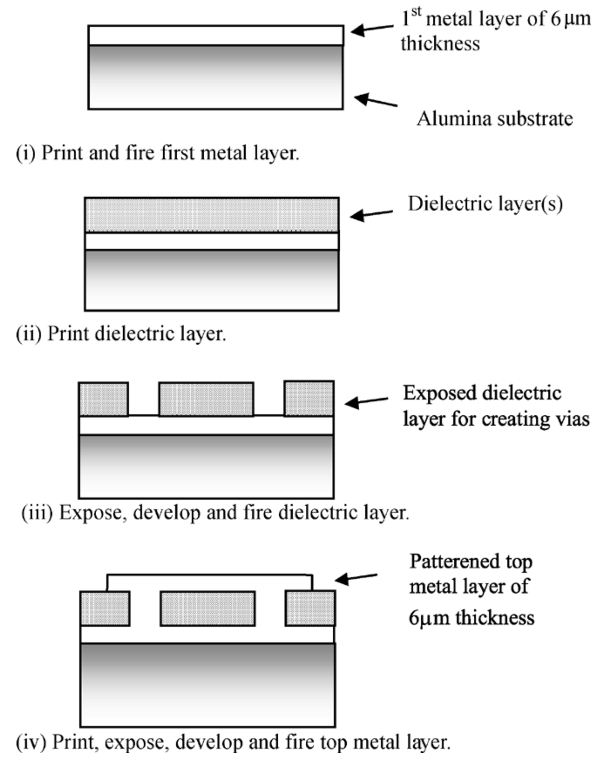

Fig. 1. Photoimageable thick-film process.

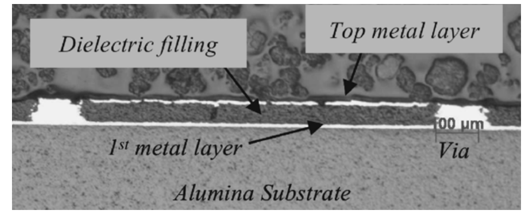

Fig. 2. Cross section of a thick-film waveguide.

TABLE I
THICK-FILM WAVEGUIDE WIDTHS FOR STANDARD BANDS

WAVEGUIDE BAND	WAVEGUIDE WIDTH (mm)
V (50 – 75 GHz)	1.34
W (75 – 110 GHz)	0.898
D (110 – 170 GHz)	0.6
G (140 – 220 GHz)	0.46

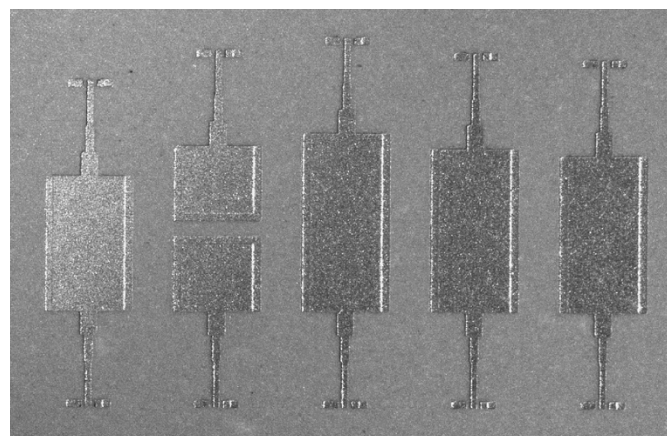

Fig. 3. Fabricated W-band waveguide–microstrip transition set for TRL calibration.

of the reduced-height waveguides, and demonstrates the level of performance that can be achieved by presenting results for SIW bandpass filters operating up to 180 GHz.

II. FABRICATION OF THICK-FILM WAVEGUIDES

Traditionally, thick-film technology has been limited to applications below 20 GHz due to the manufacturing tolerances of the process. Recently, however, the introduction of photoimageable thick-film materials has allowed the technology to produce patterns of much higher resolution, supporting the fabrication of low-loss circuits in the millimeter-wave regime. In this paper, the Hibridas HC4700 silver conductor and HD1000 dielectric ($\varepsilon_r \approx 8$) photoimageable pastes are used to fabricate millimeter-wave SIWs and resonant cavity filters.

The fabrication process is summarized in Fig. 1. First, a uniform metal layer is printed on an alumina substrate [see Fig. 1(i)]. This forms the lower broadwall of the waveguide and the ground plane for the microstrip transition. A dielectric layer is then printed [see Fig. 1(ii)] and photoimaged [see Fig. 1(iii)], forming the trenches required for the waveguide sidewalls and vias for the microstrip–coplanar waveguide (CPW) probe transition. Each dielectric print has a thickness of 10 μm (fired), and repeated prints are used to achieve the required guide height. Finally, a conductor layer is printed and photoimaged to form the upper broadwall of the waveguides and the microstrip feed lines [see Fig. 1(iv)]. After printing and imaging, each layer is dried and fired prior to the processing of subsequent layers.

Fig. 2 shows a cross section of the fabricated guide structure with the upper and lower broadwalls, and trench vias used to form the waveguide sidewalls. Each trench has a width of 100 μm. The use of trench vias as waveguide sidewalls provides complete isolation and allows for operation at higher millimeter-wave frequencies. The apparent irregularity of the sidewalls is mostly attributed to the polishing performed after the slicing of the substrate.

III. WAVEGUIDE CHARACTERISTICS AND TRANSITION DESIGN

The Hibridas dielectric paste has a nominal relative permittivity of 8. Table I shows the required waveguide width a calculated for the standard V-, W-, D-, and G-bands. To facilitate probing of the waveguide, a simple microstrip-to-waveguide transition like that proposed by Deslandes and Wu [7] is used. Use of a coplanar transition has previously been investigated [8]. However, full-wave modeling performed on both structures indicate the microstrip transition to be better suited, as it ensures minimal disruption to the field propagating between the two media. A two-section Chebyshev transformer is employed to alleviate the impedance mismatch between the waveguide and microstrip feed line. Simulations indicate this gives an improvement of 14 dB in return-loss performance compared to a direct transition from microstrip to SIW. Fig. 3 shows a microphotograph of a set of W-band back-to-back transitions designed to allow thru-reflect line (TRL) calibration.

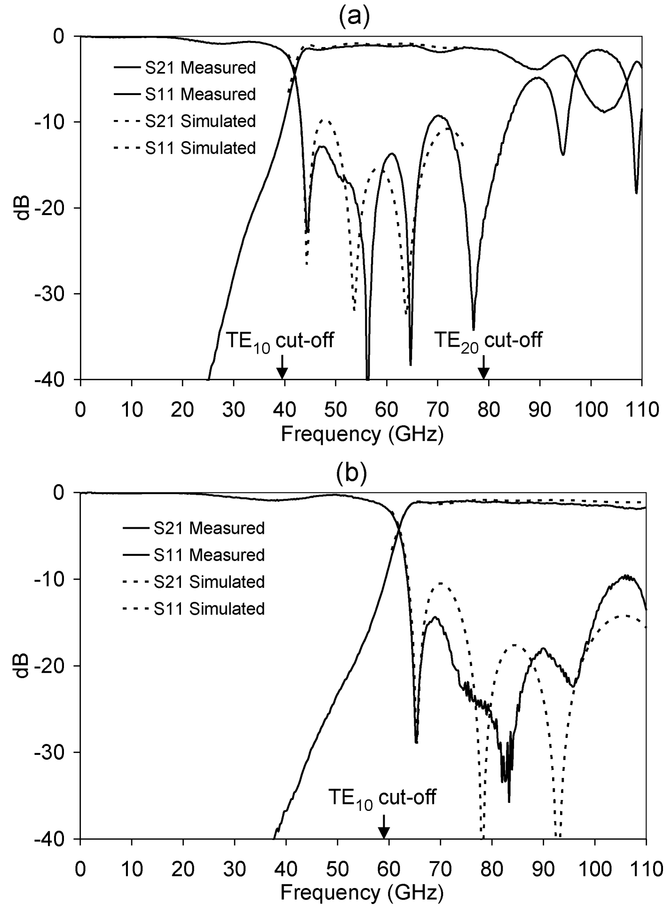

Fig. 4. Waveguide back-to-back transition measurements. (a) V-band. (b) W-band.

To first measure the characteristics of the back-to-back transitions, a broad-band line-reflect-reflect-match (LRRM) calibration using a Cascade Microtech impedance standard substrate (ISS) and Wincal was performed. As shown in Fig. 4, the measurements are in good agreement with simulated predictions, and show dominant mode propagation beginning at 39 and 59 GHz for the V- and W-band waveguides. An insertion loss of 1 dB is achieved for the V-band back-to-back transition and 1.5 dB for the W-band one, which have 6.64- and 4.35-mm waveguide lengths, respectively. Return-loss measurements indicate optimal matching at midband for both guides, as was intended with the design of the Chebyshev matching transformer. The bandwidth of the transition is found to be limited by the differing impedance-frequency characteristics of microstrip and waveguide transmission lines.

IV. Waveguide Losses

A. Loss Mechanisms

Losses in a RWG are comprised of dielectric and conductor losses. The dielectric losses of RWG can be calculated by allowing the dielectric constant to become complex. This gives an attenuation constant [22]

$$\alpha_d = \frac{\sqrt{\varepsilon_r} k_0 \tan\delta}{2\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \tag{1}$$

where f is the frequency, ε_r is the dielectric constant of the guide, k_0 is the free-space wavenumber, $\tan\delta$ is the loss tangent of the dielectric, and f_c is the TE_{10} cutoff frequency. The conductive losses of RWG can be calculated using the ideal modal currents in the waveguide walls and the surface resistivity R_s. By considering the total power flow, the conductive attenuation constant is obtained [22] as follows:

$$\alpha_c = \frac{R_s}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \sqrt{\frac{\varepsilon_r \varepsilon_0}{\mu}} \left[\frac{1}{b} + \frac{2}{a}\left(\frac{f_c}{f}\right)^2\right] \tag{2}$$

where a and b are the waveguide width and height, respectively. The total attenuation is simply the sum of the dielectric and conductive losses $\alpha = \alpha_d + \alpha_c$.

Since $f_c/f \leq 1$ and, for thick-film waveguides, $b \ll a/2$, then the second term in (2) is much less than the first. Therefore, the conduction losses are approximately inversely proportional to b as follows:

$$\alpha_c \approx \frac{R_s}{b\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \sqrt{\frac{\varepsilon_r \varepsilon_0}{\mu}}. \tag{3}$$

For a waveguide of conventional height, the loss tangent of the dielectric filling has a significant effect on the overall losses. However, since α_d is independent of b, we find that if the height b of the guide is small, as is the case in the current implementation, then the conductive losses dominate. In addition, at millimeter wavelengths, conductor losses increase significantly due to the effect of surface roughness. Conductor surface roughness is a characteristic engineered into thick-film materials to promote adhesion between layers. The processed conductor layer was found to have an rms surface roughness of 0.27 μm. Another contributing factor to conductor losses is found in currents flowing down the sidewalls of the waveguide. As the waveguide is built using a multilayer technique, misalignment of successive layers can result in an increase in current losses. However, due to the reduced height of the waveguides $(b \ll a)$, losses in the upper and lower broadwalls tend to dominate and current losses in the sidewalls do not overtly affect the loss characteristic of the waveguides.

B. Measured Results

To precisely measure the propagation characteristics of the fabricated waveguides, the probe–microstrip and the microstrip–SIW transitions are calibrated out. This is achieved by means of the multiline TRL calibration technique [23]. The calibration standards fabricated include a thru, reflect, and three line lengths, each of which correspond to $\lambda_g/4$ at the upper, lower, and middle frequencies of the respective guide bands. The chosen line lengths improve the accuracy of the algorithms used in the MultiCal software, as the calibration technique is to an extent frequency dependant. Measurements were taken on waveguides of height 30 and 60 μm for both V- and W-bands. The measurements in Fig. 5 illustrate the significant reduction in attenuation achieved by printing more layers to increase the waveguide height: In going from a 30-μm height to 60 μm, the

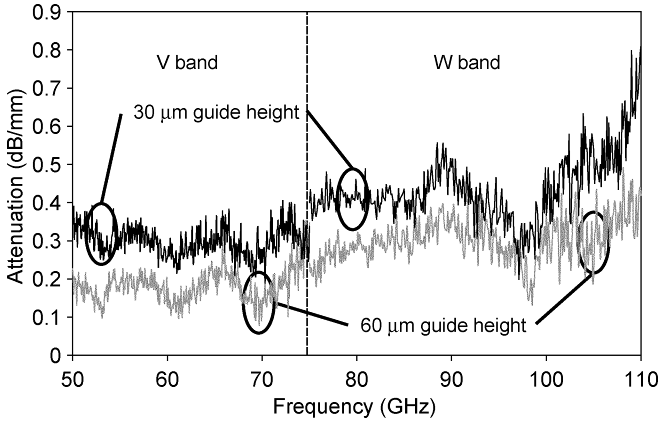

Fig. 5. Measured attenuation (in decibels per millimeter) for V- and W-band waveguides.

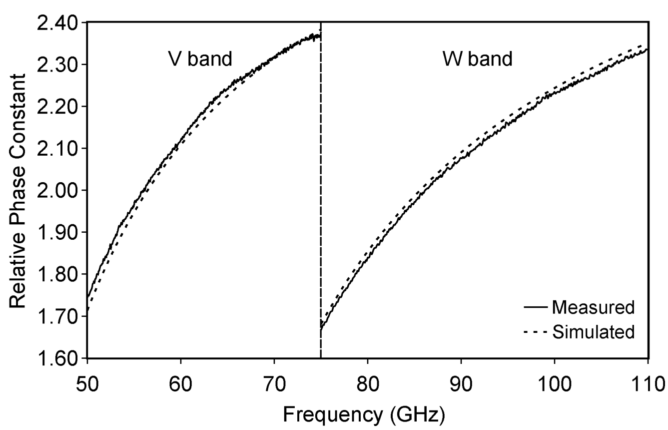

Fig. 6. Measured and calculated relative phase constant.

average loss improves from 0.32 to 0.1 dB mm^{-1} in the V-band and from 0.44 to 0.2 dB mm^{-1} in the W-band. By deducting these measured waveguide losses from the back-to-back transition data in Section III, a single microstrip–SIW transition is found to have a loss of only 0.27 dB in the V-band and 0.49 dB in the W-band.

The relative phase constant (β/k_0) for both V- and W-band waveguides is extracted from the calibrated TRL data and is in excellent agreement with theoretical predictions, as shown in Fig. 6. For the calculated values, a relative permittivity of 7.87 was used, based on previous results. The graph demonstrates that the dispersion characteristics of the photoimageable thick-film SIWs can be predicted accurately, even at millimeter wavelengths while using standard waveguide equations. Fig. 7 shows the value of ε_r extracted from the measured relative phase constant by manipulating the standard waveguide β expression to give

$$\varepsilon_r = \left(\frac{\beta}{k_0}\right)^2_{\text{meas}} + \frac{1}{k_0^2}\left(\frac{\pi}{a_{\text{meas}}}\right)^2. \tag{4}$$

The estimated uncertainty in this data is approximately $\pm$ 0.2 for ε_r based on $\pm$10-μm uncertainty in the length and width measurements.

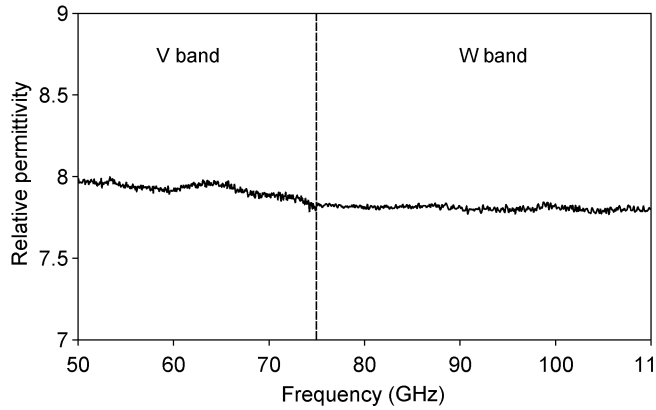

Fig. 7. Relative permittivity extracted from the measured relative phase constant versus frequency.

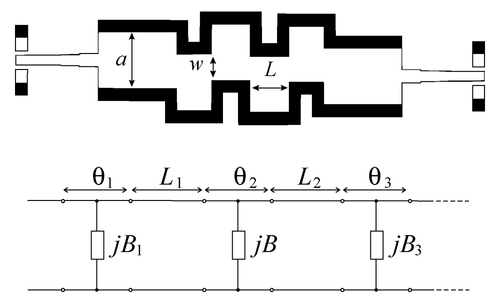

Fig. 8. H-plane offset filter with equivalent circuit.

TABLE II
DESIGN AND FABRICATED DIMENSIONS FOR THE 94-GHz FILTER

	W(mm)	*W* (FABRICATED)	*L(mm)*	*L* (FABRICATED)
1	0.536	0.52	0.6	0.59
2	0.410	0.39	0.67	0.66
3	0.382	0.36	0.67	0.66
4	0.410	0.39	0.6	0.59
5	0.536	0.52		

V. WAVEGUIDE FILTERS

A. Design

The filters presented in this paper are based on an inverter-coupled filter design [24], [25]. Series resonators coupled by appropriate shunt susceptances are used to realize a bandpass-filter characteristic. The series resonators are readily formed using sections of waveguide transmission lines, and H-plane offsets are used to realize required shunt susceptances. The H-plane offset is chosen due to its ease of manufacture using the thick-film process. The offset can be modeled as a shunt susceptance jB at the middle of a transmission line of length θ (Fig. 8). The additional length θ introduced by the offset is compensated for in the adjacent resonant cavity sections. To synthesize a Chebyshev filter response, the dimensions of the displaced waveguide junctions are calculated using the technique of Hunter [26]. The calculated values obtained for each displaced junction is further characterized using full-wave analysis by modeling the structure in HFSS. The complete

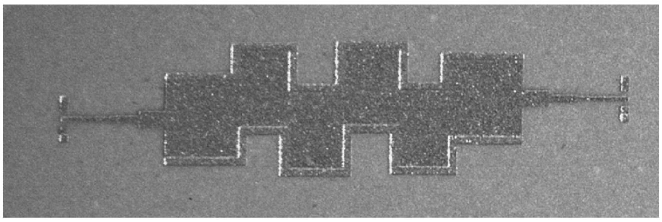

Fig. 9. Microphotograph of 94-GHz filter with RWG-to-microstrip transitions (total length: 4.34 mm).

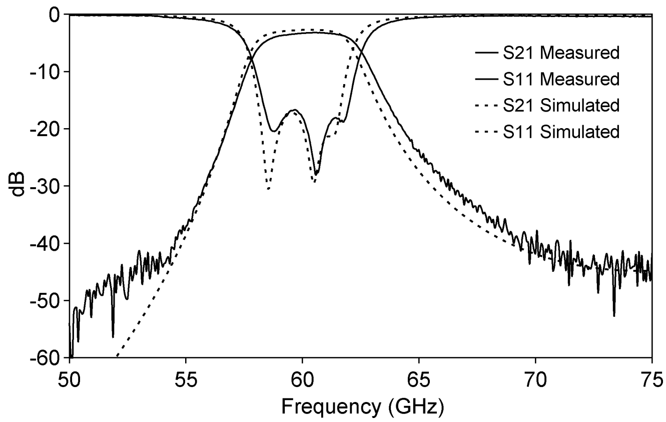

Fig. 10. Measured and modeled 60-GHz filter response.

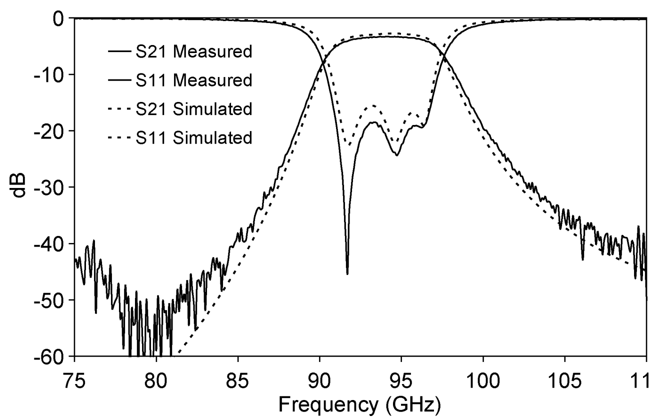

Fig. 11. Measured and modeled 94-GHz filter response.

structure is subsequently simulated to ensure the filter response is in accordance with the design specification.

Fourth-order Chebyshev filters of 0.01-dB ripple, centred at 60, 94, 150, and 180 GHz have been designed and tested. Each filter has been realized using V-, W-, D-, and G-band waveguides designed using the procedure described in Section III with a waveguide height of 60 μm. Table II shows the dimensional values for the aperture width w and length L of the cavities for the 94-GHz filter.

Fig. 9 shows a microphotograph of the 94-GHz filter. As can be seen from Table II, differences between the designed and fabricated dimensions are minimal. The process, therefore, allows highly accurate designs to be manufactured, which is essential for filter structures operating in the millimeter-wave band.

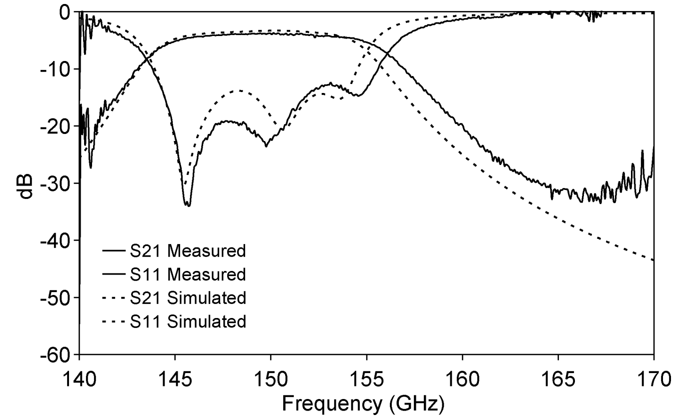

Fig. 12. Measured and modeled 150-GHz filter response.

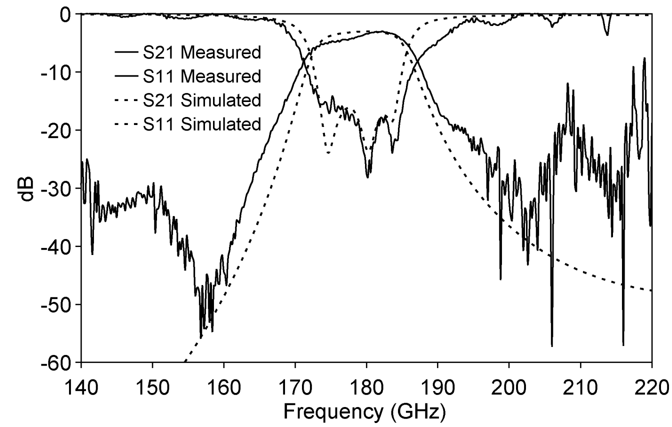

Fig. 13. Measured and modeled 180-GHz filter response.

TABLE III
FILTER PERFORMANCE

FREQUENCY (GHZ)	INSERTION LOSS (DB)	FILTER BANDWIDTH
60	-3.29	7.2%
94	-3.33	7.3%
150	-3.86	8%
180	-4.09	7%

B. Measured Results

Measurements performed on the filters used the TRL calibration and deembedding technique mentioned earlier. Removing the losses introduced by the probes and microstrip transitions gives an accurate indication of the passband insertion loss of the fabricated filters. As shown in Figs. 10–13, all the filter measurements are in good agreement with simulated predictions. The upper stopband response degradation observed in the 180-GHz filter is an artifact of the calibration procedure, and not a characteristic of the filter response. Calibration at 180 GHz was extremely challenging with the probe overtravel (skate) distance being comparable to the physical difference in length of the calibration lines. A slight deterioration in filter performance is believed to be due to the effect of dimensional tolerances, which are more significant due to the smaller operating wavelength. The insertion loss and filter bandwidth of the filters are summarized in Table III. The measured response of the filters show performance at least equal to the results obtained in [27]–[29], all of which utilized fabrication procedures far more complex than the one adopted here.

VI. Conclusion

This paper has demonstrated the potential of thick-film processing for the fabrication of hybrid circuits based on planar microstrip and miniaturized waveguide components. The loss and dispersion characteristics of the fabricated waveguide structures operating up to 110 GHz have been presented. Resonant cavity filters operating at 60, 94, 150, and 180 GHz have been designed and fabricated. The measured responses of the filter characteristics are in good agreement with predicted values, proving that the precise dimensional requirements needed for millimeter-wave circuit design can be achieved with photoimageable thick-film materials. The straightforward fabrication technique, low-loss, and highly predictable performance of the components make this an attractive technology for economical millimeter-wave system development.

Acknowledgment

The authors would like to acknowledge the use of MultiCal, which was provided by the National Institute of Standards and Technology (NIST), Boulder, CO.

References

[1] L. Yujiri, M. Schucri, and P. Moffs, "Passive millimeter-wave imaging," *IEEE Micro*, pp. 39–50, Sep. 2003.

[2] J. W. Digby, C. E. McIntosh, G. M. Parkhurst, J. W. Hadjiloucas, J. M. Chamberlain, R. D. Pollard, R. E. Miles, D. P. Steenson, N. J. Cronin, and S. R. Davies, "Fabrication and characterization of micromachined rectangular waveguide components for use at millimeter-wave and terahertz frequencies," *IEEE Trans. Microw. Theory Tech.*, vol. 48, no. 8, pp. 1293–1302, Aug. 2000.

[3] W. R. McGrath, C. Walker, M. Yap, and Y. Tai, "Silicon micromachined waveguides for millimeter-wave and submillimeter-wave frequencies," *IEEE Microw. Guided Wave Lett.*, vol. 3, no. 3, pp. 61–63, Mar. 1993.

[4] S. Lucyszyn, "Terahertz multi-chip module (T-MCM) technology for the 21st Century?," in *IEE Multi-Chip Modules RFICs Colloq.*, London, U.K., May 1998, pp. 6/1–6/8.

[5] S. Lucyszyn, D. Budimir, Q. H. Wang, and I. D. Robertson, "Design of compact monolithic dielectric-filled metal-pipe rectangular waveguides for millimeter-wave applications," *Proc. Inst. Elect. Eng.*, pt. H, vol. 143, no. 5, pp. 451–453, Oct. 1996.

[6] H. Uchimura, T. Takenoshita, and M. Fujii, "Development of a 'laminated' waveguide," *IEEE Trans. Microw. Theory Tech.*, vol. 46, no. 12, pp. 2437–2443, Dec. 1998.

[7] D. Deslandes and K. Wu, "Integrated microstrip and rectangular waveguide in planar form," *IEEE Microw. Wireless Compon. Lett.*, vol. 11, no. 2, pp. 68–70, Feb. 2001.

[8] M. S. Aftanasar, P. R. Young, I. D. Robertson, J. Minalgiene, and S. Lucyszyn, "Photoimageable thick-film millimeter-wave metal-pipe rectangular waveguides," *Electron. Lett.*, vol. 37, no. 18, pp. 1122–1123, Aug. 2001.

[9] Y. Rong, A. Zaki, J. Gipprich, M. Hageman, and D. Stevens, "LTCC wide-band ridge-waveguide bandpass filters," *IEEE Trans. Microw. Theory Tech.*, vol. 47, no. 9, pp. 1836–1840, Sep. 1999.

[10] D. Deslandes and K. Wu, "Single-substrate integration technique of planar circuits and waveguide filters," *IEEE Trans. Microw. Theory Tech.*, vol. 51, no. 2, pp. 593–596, Feb. 2003.

[11] Z. C. Hao, W. Hong, X. P. Chen, J. X. Chen, K. Wu, and T. J. Cui, "Multilayered substrate integrated waveguide (MSIW) elliptic filter," *IEEE Microw. Wireless Compon. Lett.*, vol. 15, no. 2, pp. 95–97, Feb. 2005.

[12] M. S. Aftanasar, P. R. Young, and I. D. Robertson, "Rectangular waveguide filters using photoimageable thick-film processing," in *Proc. 32nd Eur. Microwave Conf.*, Milan, Italy, Sep. 2002. [CD ROM].

[13] D. Stephens, P. R. Young, and I. D. Robertson, "Design and Characterization of 180 GHz filters in photoimageable thick film technology," *IEEE MTT-S Int. Microwave Symp. Dig.*, Jun. 2005. [CD ROM].

[14] L. Yan, W. Hong, G. Hua, J. Chen, K. Wu, and T. J. Cui, "Simulation and experiment on SIW slot array antennas," *IEEE Microw. Wireless Compon. Lett.*, vol. 14, no. 9, pp. 446–448, Sep. 2004.

[15] D. Stephens, P. R. Young, and I. D. Robertson, "W-band substrate integrated waveguide slot antenna," *Electron. Lett.*, vol. 41, no. 4, pp. 165–167, Feb. 17, 2005.

[16] W. Che, E. K. N. Yung, and K. Wu, "Millimeter-wave ferrite phase shifter in substrate integrated waveguide (SIW)," in *IEEE AP-S Int. Symp. Dig.*, vol. 4, Jun. 2003, pp. 887–890.

[17] W. D'Orazio, K. Wu, and J. Helszajn, "A substrate integrated waveguide degree-2 circulator," *IEEE Microw. Wireless Compon. Lett.*, vol. 14, no. 5, pp. 207–209, May 2004.

[18] F. Xu and K. Wu, "Guided-wave and leakage characteristics of substrate integrated waveguide," *IEEE Trans. Microw. Theory Tech.*, vol. 53, no. 1, pp. 66–73, Jan. 2005.

[19] P. Barnwell and J. Wood, "A novel thick film on ceramic MCM technology offering MCM-D performance," in *IEEE 6th Int. Multichip Modules Conf.*, Apr. 1997, pp. 48–52.

[20] B. Dziurdzia and S. Nowak, "FODEL(R) photoimageable technology in microwave applications," in *IEEE 12th Int. Microwaves and Radar Conf.*, vol. 2, May 1998, pp. 445–450.

[21] S. Muckett and J. Minalgene, "Advances in superfine line thick film materials & processing," in *Microelectronics and Packaging Int. Symp.*, 2002, pp. 20–22.

[22] S. Ramo, J. R. Whinnery, and T. van Duzer, *Fields and Waves in Communications Electronics*, 3rd ed. New York: Wiley, 1994.

[23] R. Marks, "A multiline method of network analyzer calibration," *IEEE Trans. Microw. Theory Tech.*, vol. 39, no. 7, pp. 1205–1215, Jul. 1991.

[24] G. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance Matching Networks and Coupling Structures*. Norwood, MA: Artech House, 1980.

[25] I. C. Hunter, *Theory and Design of Microwave Filters*, ser. Electromagnetic Wave. London, U.K.: IEE Press, 2001.

[26] J. D. Hunter, "The displaced rectangular waveguide junction and its use as adjustable reference reflection," *IEEE Trans. Microw. Theory Tech.*, vol. MTT-32, no. 4, pp. 387–394, Apr. 1984.

[27] S. V. Robertson, L. P. B. Katehi, and G. M. Rebeiz, "Micromachined W-band filters," *IEEE Trans. Microw. Theory Tech.*, vol. 44, no. 4, pp. 598–606, Apr. 1996.

[28] E. Rius, G. Prigent, H. Happy, G. Dambrine, S. Boret, and A. Cappy, "Wide- and narrow-band bandpass coplanar filters in the W-frequency band," *IEEE Trans. Microw. Theory Tech.*, vol. 51, no. 3, pp. 784–791, Mar. 2003.

[29] G. Pringent, E. Rius, F. L. Pennec, S. L. Maguer, C. Quendo, G. Six, and H. Happy, "Design of narrow-band DBR planar filters in Si-BCB technology for millimeter-wave applications," *IEEE Trans. Microw. Theory Tech.*, vol. 52, no. 3, pp. 1045–1051, Mar. 2004.

Daniel Stephens received the B.Eng. degree in electrical and electronic engineering from Heriot-Watt University, Edinburgh, U.K., in 2000, and the Ph.D. degree from the Advanced Technology Institute (ATI), University of Surrey, Surrey U.K, in 2005. His doctoral research focused on the development of low-cost millimeter-wave transmission lines and passive components for use in emerging technologies within the field. This concept was successfully demonstrated by the development of waveguides, filters and antennas all operating at millimeter-wave frequencies.

His current research interests include the development of low-loss filters and the accurate modeling of transmission lines.

Paul R. Young (M'00–SM'05) received the B.Eng. and Ph.D. degrees in electronic engineering from the University of Kent, Canterbury, Kent, U.K., in 1994 and 1998, respectively.

From 1997 to 1999, he was a Research Scientist with the National Physical Laboratory, Middlesex, U.K., where he developed standards for RF and microwave impedance measurement. He then spent a year as a Research Fellow with the University of Surrey before becoming a Lecturer with the University of Kent in 2001. He has authored or coauthored over 40 papers in journals and conference proceedings. His research interests include guided-wave structures, electromagnetic bandgaps, and microwave measurement techniques.

Ian D. Robertson (M'96–SM'05) received the B.Sc. (Eng.) and Ph.D. degrees from King's College London, London, U.K., in 1984 and 1990, respectively.

From 1984 to 1986, he was with the Monolithic Microwave Integrated Circuit (MMIC) Research Group, Plessey Research (Caswell) Ltd. Since then, he has held academic posts with King's College London and the University of Surrey. In June 2004, he became the Centenary Chair in Microwave and Millimeter-Wave Circuits with The University of Leeds, Leeds, U.K. He is currently the Honorary Editor of the *IEE Proceedings—Microwaves, Antennas & Propagation*. He edited *MMIC Design* (London, U.K.: IEE, 1995) and coedited *RFIC & MMIC Design and Technology* (IEE: London, U.K., 2001, 2nd ed.). He has authored or coauthored over 330 papers in the areas of MIC and MMIC design.

Dr. Robertson has organized numerous colloquia, workshops, and short courses for both the Institution of Electrical Engineers (IEE), U.K., and the IEEE.